

Contact Information

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Summary

Tech Lead with 10+ years of experience specializing in modern C++ development and team leadership. Proven track record in architecting high-performance, cross-platform systems (Linux, Windows, macOS, iOS, Android) with focus on memory optimization and concurrency. Led teams in developing production SDKs and systems software, mentoring engineers on best practices and performance-critical development. Strong expertise in technical leadership, CI/CD infrastructure, and delivering scalable solutions from concept to deployment.

Skills

Core Technical Skills

- **C++ (10+ years):** Modern C++11/14/17/20, STL, templates, multithreading, async programming with coroutines
- **Linux Development Environment:** System programming, cross-platform development, performance profiling
- **Memory Management & Performance Optimization:** Cache optimization, memory pools, profiling, memory-efficient design patterns
- **Concurrency & Multithreading:** Async programming, coroutines, synchronization primitives
- **Cross-Platform Development:** Linux, Windows, macOS, iOS, Android; low-level debugging

Leadership & Collaboration

- **Technical Leadership & Team Mentoring:** Code reviews, architectural decisions, best practices
- **Architecture Design:** System design, performance-critical development, scalable solutions
- **Communication Skills:** Cross-functional teams, documentation, technical decision-making

Development Tools & Infrastructure

- **Build Systems:** CMake, Make, Gradle
- **CI/CD & Infrastructure:** Jenkins, GitHub Actions, CI/CD pipelines, Git
- **Graphics & Rendering:** Vulkan, OpenGL, Metal, Shaders

Additional Technical Skills

- **Game Engines & Platforms:** Unity3D, Unreal Engine, Cocos2d-x, WebGL, Custom engine development
- **Scripting & Automation:** Python, Bash
- **AI & Tools:** OpenAI API, GPT-based tools, Retrieval-Augmented Generation (RAG), LlamaIndex, vector databases

Professional Experience

Tech Lead (C++), Adverty (Dec 2021–Present)

1. [AdvertySDK](#)

- **Used intensively:** C++, Metal, Vulkan, OpenGL, Linux/Windows/macOS/iOS/Android
- **Achievements:**
 - Successfully led the development and release of the SDK from scratch for cross-platform deployment (Linux, Windows, macOS, iOS, Android)
 - Led and mentored a team of 2-4 engineers, establishing coding standards and conducting code reviews

- Introduced TDD processes to the team and mentored engineers on best practices
- Championed adoption of modern C++20 features including coroutines, improving async code performance and readability
- Led the SDK architecture design with focus on performance-critical systems
- Implemented CMake-based build infrastructure with CI/CD integration (Jenkins, GitHub Actions)
- Designed the SDK API for optimal memory efficiency and cross-platform compatibility
- Built unit and integration test infrastructure
- Implemented high-performance graphics library with Vulkan/OpenGL/Metal/DX11 support, focusing on memory efficiency and cache optimization
- Developed SDK integrations for Unity3D and Unreal Engine with attention to performance on resource-constrained platforms
- Created video player with FFmpeg backend, implementing efficient memory management for multimedia streaming
- Implemented memory pool allocators and cache optimization strategies for graphics rendering
- Conducted performance profiling and optimization across multiple platforms (Linux, Windows, macOS, iOS, Android)
- Optimized memory footprint and CPU performance for resource-constrained mobile devices

Senior Software Engineer (C++), King.com (Apr 2020–Dec 2021)

1. [Pet Rescue Saga](#)

- **Used intensively:** C++, multithreading, performance optimization
- **Achievements:**
 - Architected high-performance game engine using ECS and data-oriented design, optimizing cache locality and memory access patterns
 - Improved runtime performance through memory layout optimization and cache-friendly data structures
 - Created debugging tools using the ImGui library
 - Led codebase refactoring efforts, improving code maintainability and performance across the codebase

Senior Software Engineer (C++), Vizion Games (Jun 2019–Apr 2020)

1. [Mahjong Treasure Quest](#)

- **Used intensively:** C++, Cocos2d-x, cross-platform development
- **Achievements:**
 - Developed memory-efficient UI system with custom CSS parser for cross-platform rendering
 - Implemented object transformations (transform/scale/rotate) with performance in mind
 - Significantly improved developer tools using ImGui, reducing debugging time and enhancing team productivity
 - Designed and implemented a new Flow Actions System with modular architecture

Senior Software Engineer (C++/C#), Gambino (2015–2019)

1. [Slototron.com](#)

- **Used intensively:** C++, Emscripten, JavaScript, multithreading, request handling
- **Achievements:**
 - Developed secure payment system with C++/JavaScript interop, demonstrating cross-language communication skills
 - Implemented request retry mechanism with concurrency handling, significantly improving client-server reliability
 - Optimized application performance through request batching, memory caching, and resource management strategies
 - Built a runnable prototype using C++/Emscripten for Instant Games, demonstrating cross-platform capability

2. [Gambino Slots](#)

- **Used intensively:** C#, Xamarin, Dart
- **Achievements:**
 - Worked on a C# to Dart converter based on the Roslyn compiler

- Rewrote the game in Dart from scratch
- Ported the in-house C# game engine to macOS
- Integrated OpenGL's scissor mechanism
- Integrated Firebase push messaging
- Refactored and modularized the codebase per platform

Earlier Software Engineering Experience (2013–2016)

1. Multiple Roles (C#), MuGames, Gambino, Playtika

- **Used intensively:** C#, Xamarin, Autofac, in-house game engines
- **Highlights:**
 - Developed mobile casino and social gaming applications using C#, Xamarin
 - Implemented CI/CD pipelines, ad management systems (Vungle, Chartboost, Unity), and social platform integrations (Facebook, Viber)
 - Built testing infrastructure and anti-cheating mechanisms
 - Worked with multithreading, dependency injection, and cross-platform development
 - Specific projects: WildRide (ad manager, social plugin system), Viber Rude Rider (Viber integration), Caesars Casino (slot machines, sound manager, reels engine), Gambino Slots (slot infrastructure, cheat/debug systems)